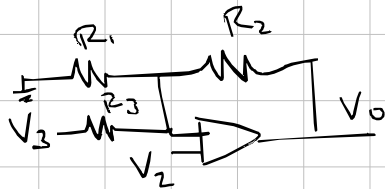


D_1 has for. drop V_f ...

if virtual short is active, then $V_1 = V_2$

two cases:

- ① $V_1 \geq V_2 + V_f$; virtual short is maintained and effectively none!



$$V_3 = V_1 - V_f$$

Superposition

V_2 : ($V_3 = 0$) $\rightarrow$ non-inv. Amp! $V_{O2} = \left(1 + \frac{R_2}{R_{eff}}\right) V_2$

$$R_{eff} = R_1 \parallel R_3 = \frac{R_1 R_3}{R_1 + R_3} \rightarrow V_{O2} = \left(1 + \frac{R_2 (R_1 + R_3)}{R_1 R_3}\right) V_2$$

V_3 : ($V_2 = 0$) $\rightarrow$ No current through R_1 (ground on both sides)
inv. Amp! $V_{O3} = -\frac{R_2}{R_3} V_3$

$$V_o = \left(1 + \frac{R_2 (R_1 + R_3)}{R_1 R_3}\right) V_2 - \frac{R_2}{R_3} V_3$$

if $R_1 = R_2 = R_3 \neq R_0$

$$V_o = \left(1 + \frac{R(2R)}{R^2}\right) V_2 - V_3 = (1 + 2) V_2 - V_3 = 3V_2 - V_3$$

Case ②: $V_3 \leq V_2$, $V_1 \leq V_2 + V_f$

D_1 off but feedback still exists $\rightarrow$ virtual short still happening!

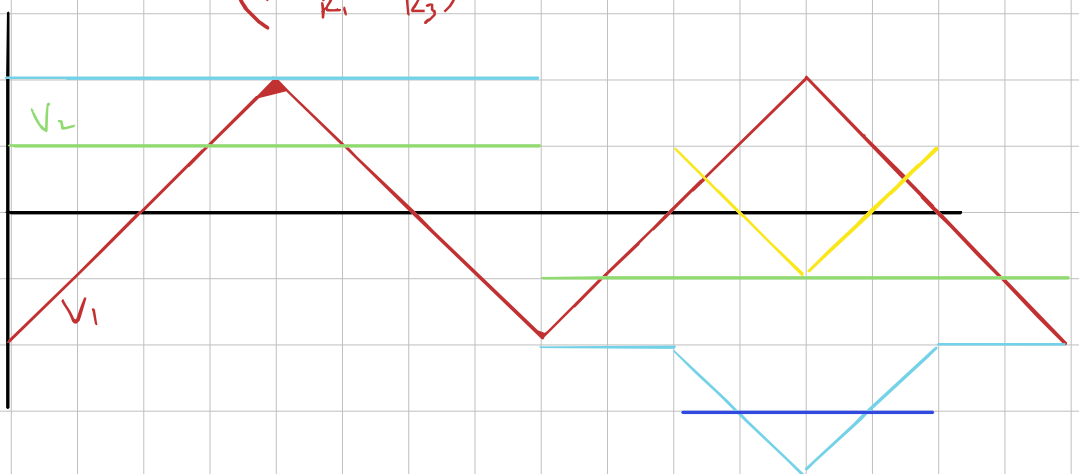


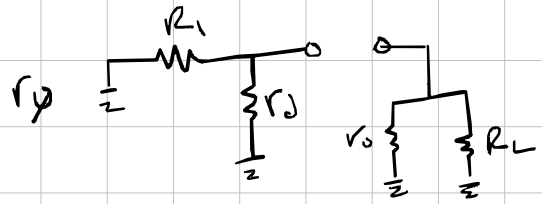
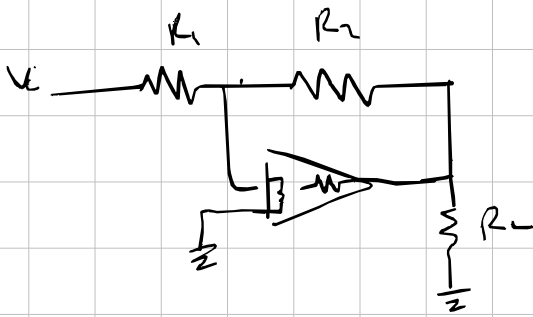
Non-inv. Amp!

$$V_o = \left(1 + \frac{R_2}{R_1}\right) V_2$$

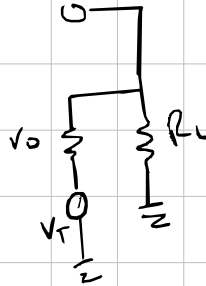
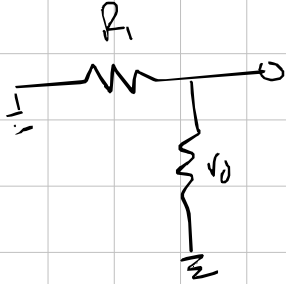
$$V_o = \begin{cases} \left(1 + \frac{R_2}{R_3}\right) V_2 & V_1 \leq V_2 + V_f \\ \left(1 + \frac{R_2 (R_1 + R_3)}{R_1 R_3}\right) V_2 - \frac{R_2}{R_3} (V_3 - V_f) & V_1 > V_2 + V_f \end{cases}$$

$\left(1 + \frac{R_2}{R_1} - \frac{R_2}{R_3}\right)$





$$r_p = R_1 \parallel r_o + R_L \parallel r_o$$

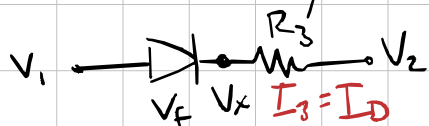


$$T_{DC}: V_n = V_p = 0 \rightarrow T_{DC} = 0$$

$$T_{AC}: V_o = \frac{R_1 \parallel r_o \parallel R_L}{R_1 \parallel r_o \parallel R_L + r_o} = V_n$$

② $f_L = 100 \text{ mHz}$ $f_H = 100 \text{ kHz}$
 $V_1 = 3 \text{ V}$ $V_2 = -3 \text{ V}$ $T = 25^\circ \text{C} = 298 \text{ K}$
 $R_1 = 10 \text{ k}\Omega$, $R_2 = 20 \text{ k}\Omega$, $R_3 = 5 \text{ k}\Omega$
 $V_f = 0.7 \text{ V}$ diode $\bar{i}_n^L = 2qI_D$ $q = 1.603 \times 10^{-19} \text{ C}$
 op Amp: $e_{nw} = 30 \text{ nV}/\sqrt{\text{Hz}}$, $f_{ce} = 50 \text{ Hz}$, $i_{nw} = 100 \text{ pA}/\sqrt{\text{Hz}}$, $f_i = 0$
 $f_B \gg 100 \text{ kHz}$ $\leftarrow$ don't have to worry about noise decay do to amp pole(s).

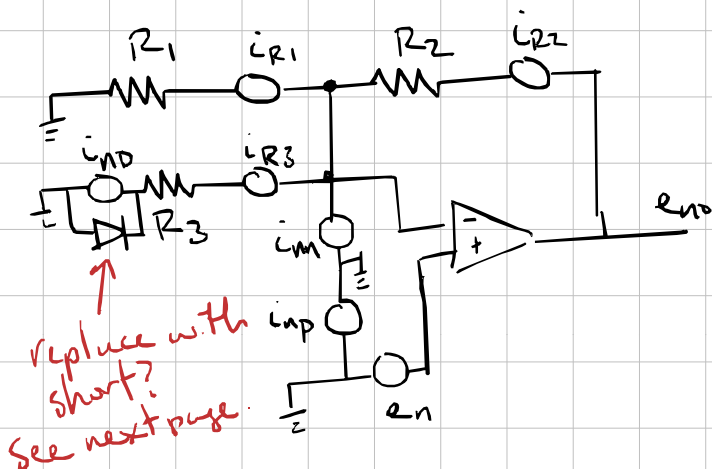
Need current through L diode: easy with virtual short:



$V_x = 2.3 \text{ V}$ $I_D = I_3 = \frac{V_x - V_2}{R_3} = \frac{5.3 \text{ V}}{5 \text{ k}\Omega} = 1.06 \text{ mA}$
 Diode $\bar{i}_n^L = 3.4 \times 10^{-22} \text{ A}/\sqrt{\text{Hz}}$

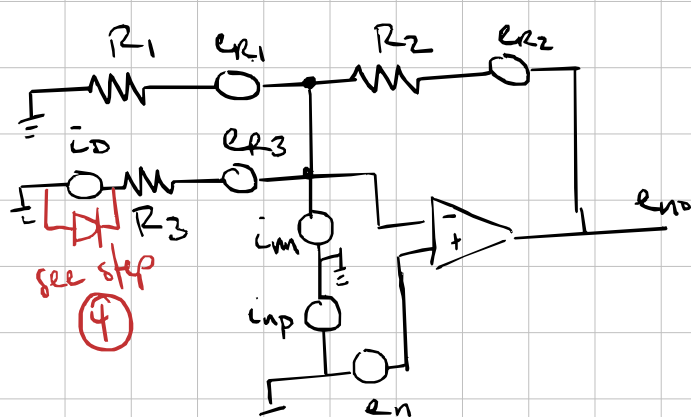
Rest of analysis: V_1 & V_2 don't matter ground/ignore.

we can do this because we know D_1 is on
 & therefore we can linearize & apply superposition
 by removing the diode.



Two methods: Direct or shortcut to e_{no} .
 Both done on following pages

② Continued: Direct calculation of e_{no} :



(Assuming Diode is on $\rightarrow$ path to gnd or other DC voltage)

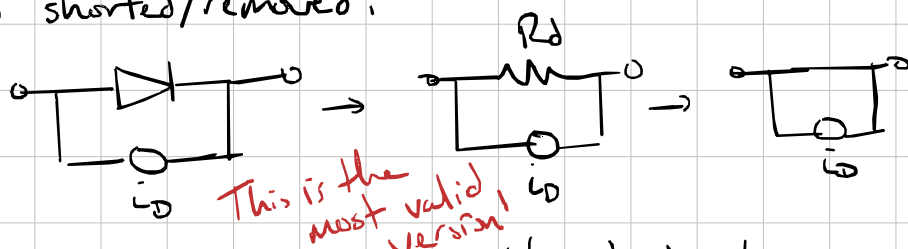
Superposition:

① e_{R1} : $\rightarrow e_{no1} = -\frac{R_2}{R_1} e_{R1}$

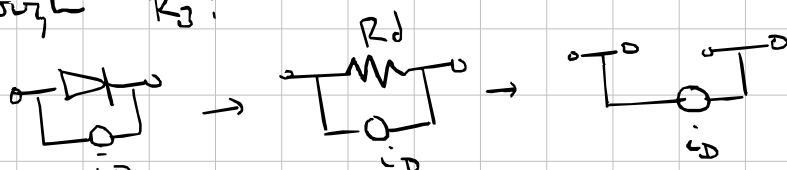
② e_{R2} : $\rightarrow e_{no2} = -\frac{R_2}{R_3} e_{R3}$

③ e_{R3} : $\rightarrow e_{no3} = i_{R2}$

④ i_D : Problem! if we assumed D_1 is perfectly on, then its on Resistance $= 0$ & the current noise is shorted/removed!



Otherwise we may assume the diode has a high on resistance (not realistic) such that " R_d " $\gg R_3$ and all noise current goes through R_3 :



I will accept either for this question! since the first is trivial ($e_{no}|_{i_D=0} = 0$) I will do the second...

$V_i = i_D R_3 \rightarrow e_{no4} = -\frac{R_2}{R_3} i_D R_3$
 $e_{no4} = R_2 i_D$

⑤ i_{nn} : current must go through R_2 :
 $e_{no5} = R_2 i_{nn}$

⑥ i_{np} : Grounded! $e_{no6} = 0$

⑦ e_n : $e_{no7} = \left(1 + \frac{R_2}{R_1 || R_3}\right) e_n$
 A_n

$$e_{no}^2 = \left(\frac{R_2}{R_1} e_{R1}\right)^2 + \left(\frac{R_2}{R_3} e_{R3}\right)^2 + e_{R2}^2 + R_2^2 (i_D^2 + i_{nn}^2) + \left[\left(1 + \frac{R_2}{R_1 || R_3}\right) e_n\right]^2$$

inserting #s & evaluating: ($e_R^2 = 4kTR$)

$$e_{no}^2 = 6.56 \times 10^{-16} \left(\frac{V^2}{Hz}\right) + 1.32 \times 10^{-16} + 3.29 \times 10^{-16}$$

$$+ 20000^2 \left(\underbrace{3.4 \times 10^{-22}}_{\text{this is } i_D. \text{ Note that it is negligible anyway!}} + 1 \times 10^{-20} \right) + 4.4 \times 10^{-14} \left(\frac{50}{f} + 1 \right)$$

$$e_{no}^2 = 4 \times 10^{-12} + 4.4 \times 10^{-14} \left(\frac{50}{f} + 1 \right)$$

woops, current noise from amp dominates!

$$e_{no}^2 = \frac{2.2 \times 10^{-12}}{f} + 4 \times 10^{-12}$$

$$e_{ni}^2 = e_{no}^2 / A_n^2 \quad A_n = \left(1 + \frac{R_2}{R_1 || R_3}\right) = 7$$

$$e_{ni}^2 = \frac{4.5 \times 10^{-14}}{f} + 8.16 \times 10^{-14}$$

(take $\sqrt{\quad}$ for actual answers)

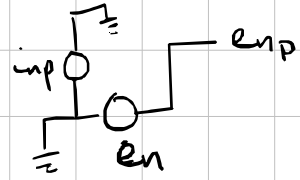
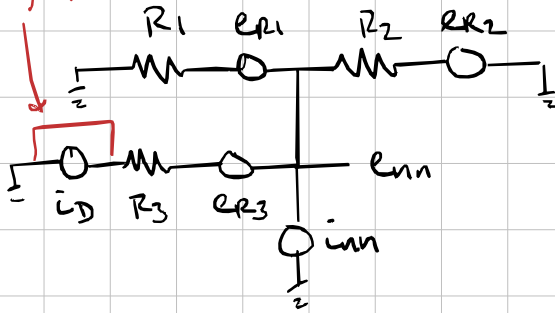
$$E_{no}^2 = \int_{f_L}^{f_H} e_{ni}^2 df = 2.2 \times 10^{-12} \ln\left(\frac{f_H}{f_L}\right) + 4 \times 10^{-12} (f_H + f_L)$$

$$3.04 \times 10^{-11} + 4 \times 10^{-7}$$

$$E_{no}^2 \approx 4 \times 10^{-7} \rightarrow E_{no} = 0.63 \text{ mV}$$

② cont. e_{ni} shortcut: (id using same justification as previous page $\rightarrow$ step 4)

see previous page



$$e_{np} = e_n$$

e_{ni} :

① e_{R1} : $e_{nn1} = e_{R1} \left(\frac{R_2 || R_3}{R_2 || R_3 + R_1} \right) \quad e_{nn1}^2 = 1.34 \times 10^{-17}$

② e_{R2} : $e_{nn2} = e_{R2} \left(\frac{R_1 || R_3}{R_1 || R_3 + R_2} \right) \quad e_{nn2}^2 = 6.7 \times 10^{-18}$

③ e_{R3} : $e_{nn3} = e_{R3} \left(\frac{R_1 || R_2}{R_1 || R_2 + R_3} \right) \quad e_{nn3}^2 = 2.7 \times 10^{-17}$

④ i_D (if short didn't exist): $e_{nn4} = (R_1 || R_2) i_D \quad e_{nn4}^2 = 1.51 \times 10^{-14}$

⑤ i_{nn} : $e_{nn5} = (R_1 || R_2 || R_3) i_{nn} \quad e_{nn5}^2 = 8.16 \times 10^{-14}$

$$e_{nn}^2 = 9.7 \times 10^{-14} \text{ V}^2/\text{Hz} \quad e_{np}^2 = 9 \times 10^{-16} \left(\frac{50}{f} + 1 \right)$$

$$e_{ni}^2 = \frac{4.5 \times 10^{-14}}{f} + 9.7 \times 10^{-14}$$

different from previous answer as i didn't neglect i_D in simplification

$$e_{no}^2 = A_n^2 e_{ni}^2 = 49 e_{ni}^2$$

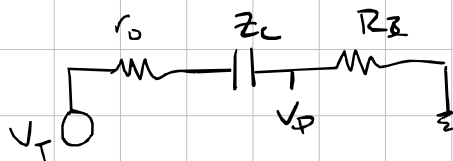
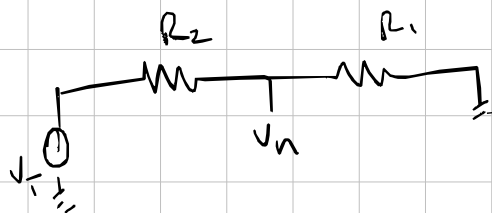
$$e_{no}^2 = \frac{2.2 \times 10^{-12}}{f} + 4.8 \times 10^{-12}$$

$$E_{no}^2 = \int_{f_L}^{f_H} e_{no}^2 df = 2.2 \times 10^{-12} \ln(f_H/f_L) + 4.8 \times 10^{-12} (f_H - f_L)$$

$3.04 \times 10^{-11} \quad 4.8 \times 10^{-7}$

$$E_{no}^2 \approx 4.8 \times 10^{-7}$$

$$E_{no} \approx 0.69 \text{ mV}$$



$$V_n = \frac{R_1}{R_1 + R_2} V_i$$

$$V_p = \frac{R_3}{R_3 + r_o + Z_c} V_i$$

$$V_p \rightarrow \frac{j\omega R_3 C}{1 + j\omega (R_3 + r_o) C} V_i$$

$$\beta = - \frac{V_p - V_n}{V_i} = - \frac{R_1}{R_1 + R_2} + \frac{j\omega R_3 C}{1 + j\omega (R_3 + r_o) C}$$

$$= \frac{R_1}{R_1 + R_2} \left[-1 + \frac{\frac{R_1 + R_2}{R_1} j\omega R_3 C}{1 + j\omega (R_3 + r_o) C} \right]$$

$$= \frac{R_1}{R_1 + R_2} \left[\frac{-1 - j\omega (R_3 + r_o) C}{1 + j\omega (R_3 + r_o) C} + \frac{\frac{R_1 + R_2}{R_1} j\omega R_3 C}{1 + j\omega (R_3 + r_o) C} \right]$$

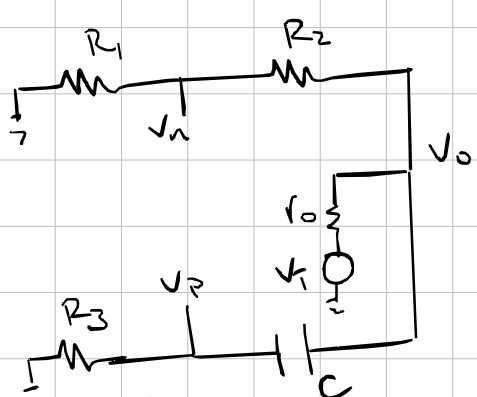
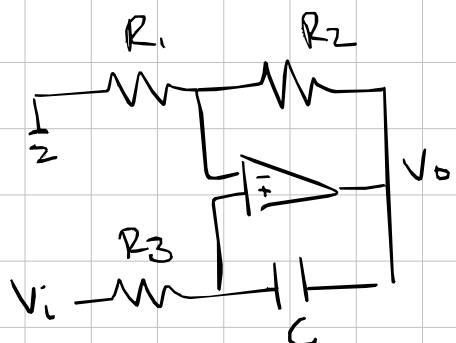
Denominator matches

Numerator

$$-1 - \cancel{j\omega (R_3 + r_o) C} + \cancel{j\omega R_3 C} + j\omega \frac{R_2 R_3}{R_1} C$$

$$-1 - j\omega r_o C + j\omega \frac{R_2 R_3}{R_1}$$

3)



$$V_o = \frac{(R_1 + R_2) \parallel (R_3 + \frac{1}{j\omega C})}{(R_1 + R_2) \parallel (R_3 + \frac{1}{j\omega C}) + r_o} V_T$$

$$V_n = \frac{R_1}{R_1 + R_2} V_o$$

$$V_p = \frac{R_3}{R_3 + 1/j\omega C} V_o = \frac{j\omega R_3 C}{1 + j\omega R_3 C} V_o$$

$$V_o = \left[\frac{j\omega R_3 C}{1 + j\omega R_3 C} - \frac{R_1}{R_1 + R_2} \right] V_o$$

$$= \left[\frac{j\omega R_3 C}{1 + j\omega R_3 C} - \frac{R_1}{R_1 + R_2} \right] \left[\frac{(R_1 + R_2) \parallel (R_3 + 1/j\omega C)}{(R_1 + R_2) \parallel (R_3 + 1/j\omega C) + r_o} \right] V_T$$

$$\frac{(R_1 + R_2)(R_3 + 1/j\omega C)}{R_1 + R_2 + R_3 + 1/j\omega C}$$

$$= \frac{R_1 + R_2 + j\omega (R_1 + R_2) R_3 C}{1 + j\omega (R_1 + R_2 + R_3) C}$$

$$= (R_1 + R_2) \frac{1 + j\omega R_3 C}{1 + j\omega (R_1 + R_2 + R_3) C}$$

$$\rightarrow (R_1 + R_2) \frac{1 + j\omega R_3 C}{1 + j\omega (R_1 + R_2 + R_3) C}$$

$$\frac{(R_1 + R_2) \frac{1 + j\omega R_3 C}{1 + j\omega (R_1 + R_2 + R_3) C} + r_o}{(R_1 + R_2) \frac{1 + j\omega R_3 C}{1 + j\omega (R_1 + R_2 + R_3) C} + r_o}$$

$$= \frac{1 + j\omega R_3 C}{1 + j\omega R_3 C + \frac{r_o}{R_1 + R_2} [1 + j\omega (R_1 + R_2 + R_3) C]}$$

$$= \frac{1 + j\omega R_3 C}{1 + \frac{r_o}{R_1 + R_2} + j\omega \left[R_3 + \frac{r_o}{R_1 + R_2} (R_1 + R_2 + R_3) \right] C}$$

$$= \frac{1 + j\omega R_3 C}{1 + \frac{r_o}{R_1 + R_2} + j\omega \left[R_3 + r_o \left(1 + \frac{R_3}{R_1 + R_2} \right) \right] C}$$

$$= \frac{1 + j\omega R_3 C}{1 + \frac{r_o}{R_1 + R_2} + j\omega \left[R_3 \left(1 + \frac{r_o}{R_1 + R_2} \right) + r_o \right] C}$$

$$\text{if } r_o \ll R_1 + R_2, \text{ then } \frac{r_o}{R_1 + R_2} \ll 1$$

$$\approx \frac{1 + j\omega R_3 C}{1 + j\omega (R_3 + r_o) C}$$

$$\beta = - \left[\frac{j\omega R_3 C}{1 + j\omega R_3 C} - \frac{R_1}{R_1 + R_2} \right] \left(\frac{1 + j\omega R_3 C}{1 + j\omega (R_3 + r_o) C} \right)$$

getting in terms of 1s & zeros

$$\beta = \frac{R_1}{R_1 + R_2} \left[- \left(\frac{R_1 + R_2}{R_1} \right) \frac{j\omega R_3 C}{1 + j\omega R_3 C} + \frac{1 + j\omega R_3 C}{1 + j\omega R_3 C} \right] \left[\frac{1 + j\omega R_3 C}{1 + j\omega (R_3 + r_o) C} \right]$$

$$= \frac{1 + j\omega \left(1 - \frac{R_1 + R_2}{R_1} \right) R_3 C}{1 + j\omega R_3 C} = \frac{1 + j\omega \frac{R_2}{R_1} R_3 C}{1 + j\omega R_3 C}$$

$$\beta = \frac{R_1}{R_1 + R_2} \frac{1 + j\omega \left(-\frac{R_2}{R_1} \right) R_3 C}{1 + j\omega R_3 C} \frac{1 + j\omega R_3 C}{1 + j\omega (R_3 + r_o) C}$$

$$\beta = \frac{R_1}{R_1 + R_2} \frac{1 + j\omega \left(-\frac{R_2}{R_1} \right) R_3 C}{1 + j\omega (R_3 + r_o) C}$$

$$\omega_p = \frac{R_1}{R_2 R_3 C} \Rightarrow$$

$$\omega_z = 1 / (R_3 + r_o) C$$

limits:

$f \rightarrow 0$

$$\beta \approx \frac{R_1}{R_1 + R_2}$$

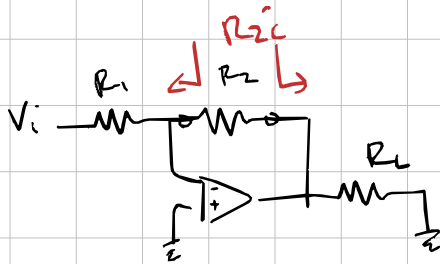
$f \rightarrow \infty$

$$\beta \approx - \frac{R_1}{R_1 + R_2} \left(\frac{R_2 / R_1 R_3 C}{(R_3 + r_o) C} \right)$$

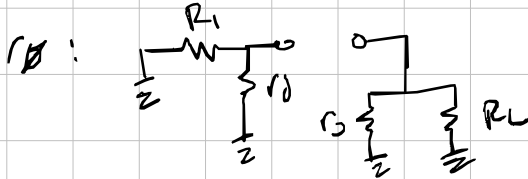
$$\beta \approx - \frac{R_1}{R_1 + R_2} \frac{R_2}{R_1} \frac{R_3}{R_3 + r_o}$$

$$\beta \approx - \frac{R_2}{R_1 + R_2} \frac{R_3}{R_3 + r_o}$$

④

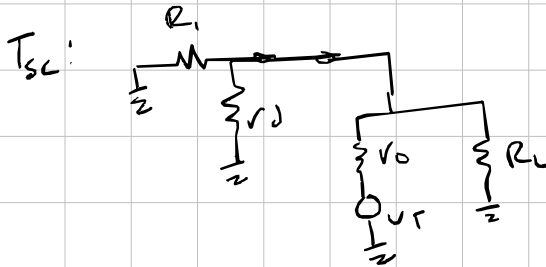


$\beta \neq 0, r_o \neq 0$



$$r_o = R_1 \parallel r_d + r_o \parallel R_L$$

$$r_o = \frac{R_1 r_d}{R_1 + r_d} + \frac{r_o R_L}{r_o + R_L}$$



$$V_P = 0$$

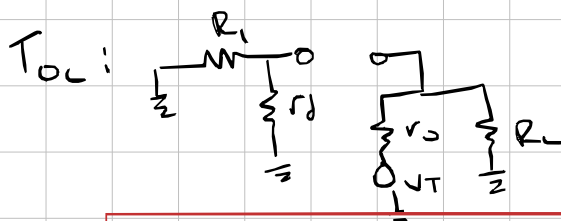
$$V_N = V_T$$

$$\frac{R_1 \parallel r_d \parallel R_L}{R_1 \parallel r_d \parallel R_L + r_o} = \frac{R_1 R_L r_d}{R_1 R_L + R_1 r_d + R_L r_d}$$

Let $X = R_1 R_L + R_1 r_d + R_L r_d$

$$V_N = V_T \frac{R_1 R_L r_d}{R_1 R_L r_d + X r_o}$$

$$T_{sc} = -a \frac{(V_P - V_N)}{V_T} = \frac{R_1 R_L r_d}{R_1 R_L r_d + X r_o}$$



$$V_N = V_P = 0$$

$$T_{oc} = 0$$

$$R_{zi} = \left[\frac{R_1 r_d}{R_1 + r_d} + \frac{r_o R_L}{r_o + R_L} \right] \left(1 + a \frac{R_1 R_L r_d}{R_1 R_L r_d + X r_o} \right)$$

Not needed: if $r_d \rightarrow \infty, r_o \rightarrow 0$, then:

$$R_{zi} = \left[\frac{R_1 r_d}{r_d} + \frac{r_o R_L}{R_L} \right] \left(1 + a \frac{R_1 R_L r_d}{R_1 R_L r_d} \right)$$

Big small

$$R_{zi} = R_1 (1 + a)$$